



503A & 503B Compounded Rx

Product Information, Nature Classification & Side Effect Reference

NATURAL

Natural/Bioidentical — identical or derived from naturally occurring compounds

SEMI-SYN

Semi-Synthetic — structurally modified analog of a natural compound

SYNTHETIC

Synthetic — fully laboratory-synthesized; no direct natural counterpart

■ **Side Effects Column** — Lists the most commonly reported adverse effects. This is a clinical reference, not an exhaustive list. Individual responses vary. Always conduct a full patient assessment before prescribing.

Understanding 503A & 503B Compounding

503A — Traditional Patient-Specific

Requires a valid **individual patient prescription**. Regulated by state pharmacy boards. Must comply with USP <795>/<797> standards.

- Patient-specific Rx required
- State pharmacy board oversight
- Custom dose & formulation
- No FDA batch approval

503B — FDA-Registered Outsourcing

Federally registered under **FDA oversight**. May produce large batches for healthcare facilities without individual Rx. Operates under CGMP standards.

- FDA-registered & inspected
- CGMP manufacturing standards
- Can supply facilities without Rx
- Consistent batch quality controls

GLP-1 & Metabolic Agents

Semaglutide and Tirzepatide are fully synthetic incretin mimetics. Most side effects are GI-related and dose-dependent, typically improving with titration.

Product	Conc	Vial	Nature	Common Side Effects	Clinical Notes
Semaglutide + Glycine (2.5mg)	2.5mg/5mg/mL	1 mL	SYNTHETIC	Nausea, vomiting, diarrhea, constipation, injection site reaction	Starter dose; weekly SC. Glycine is a natural amino acid
Semaglutide + Glycine (5mg)	2.5mg/5mg/mL	2 mL	SYNTHETIC	Nausea, fatigue, reduced appetite, GI discomfort	Standard maintenance dose
Semaglutide + Glycine (12.5mg)	2.5mg/5mg/mL	5 mL	SYNTHETIC	GI upset, pancreatitis risk (rare), headache	Higher-dose / longer supply
Semaglutide + B12 (2.5mg)	1.25mg/0.5mg/mL	2 mL	SYNTHETIC	Nausea, injection site pain; B12 generally well tolerated	B12 supports energy & tolerability
Semaglutide + B12 (5mg)	2.5mg/0.5mg/mL	2 mL	SYNTHETIC	GI symptoms, decreased appetite, mild dizziness	Common titration step
Semaglutide + B12 (12.5mg)	3.125mg/0.5mg/mL	4 mL	SYNTHETIC	Nausea, vomiting, gallbladder issues (long-term)	Extended supply
Tirzepatide (18mg)	9mg/mL	2 mL	SYNTHETIC	Nausea, diarrhea, vomiting, decreased appetite, fatigue	Dual GIP/GLP-1; potent metabolic
Tirzepatide (36mg)	18mg/mL	2 mL	SYNTHETIC	GI upset, injection site reactions, hypoglycemia risk	Mid-range maintenance
Tirzepatide (72mg)	18mg/mL	4 mL	SYNTHETIC	Nausea, constipation, pancreatitis risk (rare), hair loss	High-dose / 4-week supply

Product	Conc	Vial	Nature	Common Side Effects	Clinical Notes
Tirzepatide + B12 (17mg)	8.5mg/0.5mg/mL	2 mL	SYNTHETIC	GI symptoms, injection site irritation	Enhanced energy support
Tirzepatide + B12 (34mg)	17mg/0.5mg/mL	2 mL	SYNTHETIC	Nausea, diarrhea, reduced appetite	Mid-dose with B12
Tirzepatide + B12 (68mg)	17mg/0.5mg/mL	4 mL	SYNTHETIC	GI effects, gallbladder disease risk (long-term)	High-dose long-acting

Sublingual Formulations

Needle-free delivery. GI side effects may be milder than injectable forms due to lower peak plasma concentrations.

Product	Strength	Vol	Nature	Common Side Effects	Benefit Highlight
Sublingual Tirzepatide + NAD+	5–17.5mg/100mg	8 mL	SYNTHETIC	Nausea, oral mucosal irritation, mild flushing (NAD+)	Synthetic GLP-1 + bioidentical NAD+ (6 strengths)
Sublingual Semaglutide	2–10mg	8 mL	SYNTHETIC	Nausea, taste disturbance, dry mouth, mild GI upset	Needle-free GLP-1; 5 strengths
Sublingual Tirzepatide	5–30mg	8 mL	SYNTHETIC	Nausea, reduced appetite, oral irritation, headache	Needle-free dual agonist; 6 strengths

Peptide & Regenerative Agents

Peptides range from bioidentical to fully synthetic. Side effects are generally mild; most commonly injection site reactions and transient GI or neurological symptoms.

Peptide	Concentration	Nature	Common Side Effects	Primary Action
AOD-9604	1.2mg/mL • 5mL	SEMI-SYN	Injection site redness, nausea (rare), headache	Modified GH fragment; fat lipolysis; no IGF-1 effects
BPC-157	3mg/mL • 5mL	SEMI-SYN	Mild nausea, dizziness, transient fatigue	Gut & tissue repair; anti-inflammatory; tendon healing
DSIP	1mg/mL • 5mL	NATURAL	Excessive drowsiness, headache, mild hypotension	Natural brain peptide; sleep architecture & stress
EPITHALON	2mg/mL • 5mL	SYNTHETIC	Injection site reactions, transient fatigue, vivid dreams	Telomerase activation; anti-aging; pineal regulation
GHK-Cu	10mg/mL • 5mL	NATURAL	Skin flushing (topical), mild injection site irritation	Copper peptide; collagen synthesis; wound & skin healing
GONADORELIN	1mg/mL • 5mL	NATURAL	Hot flashes, headache, mood changes, testicular ache	Bioidentical GnRH; LH/FSH support; TRT adjunct
IGF-LR3	200mcg/mL • 5mL	SEMI-SYN	Hypoglycemia, headache, joint pain, fluid retention	Long-arg3 analog; muscle growth; anabolic signaling
Kisspeptin	1mg/mL • 5mL	NATURAL	Flushing, mild GnRH-related hormonal shifts	Endogenous neuropeptide; reproductive hormone regulation
MOTS-C	2mg/mL • 5mL	NATURAL	Injection site reactions, mild GI upset (rare)	Mitochondrial peptide; metabolic flexibility; exercise
NAD+	100mg/mL • 10mL	NATURAL	Flushing, nausea, headache, chest tightness (IV rapid)	Bioidentical coenzyme; cellular energy & DNA repair
PT-141	2mg/mL • 5mL	SEMI-SYN	Nausea, flushing, headache, transient hypertension	Melanocortin analog; CNS-mediated libido enhancement
Sermorelin	3mg/mL • 5mL	SEMI-SYN	Injection site pain, flushing, dizziness, headache	Synthetic GHRH analog (1-29); stimulates GH release

Peptide	Concentration	Nature	Common Side Effects	Primary Action
Tesamorelin	3mg/mL • 5mL	SEMI-SYN	Injection site reactions, fluid retention, joint pain	FDA-approved GHRH; visceral fat reduction; IGF-1 stimulation
Thymosin A-1	1mg/mL • 5mL	NATURAL	Generally well tolerated; mild injection site reactions	
Lipo-B	50/50/25/1mg/mL	SYNTHETIC	Injection site pain, urinary odor, mild GI effects	MIC + B12; lipotropic; liver detox support
Glutathione	200mg/mL • 10mL	NATURAL	Skin lightening (with frequent use), mild GI, cramping	Master antioxidant; detoxification; skin brightening

Combination Peptide Protocols

Side effects reflect the combined profile of each component. Additive GI or CNS effects are possible. Start at lower frequencies when introducing combination protocols.

Combination	Dose/mL	Nature	Common Side Effects	Synergistic Goal
AOD-9604 + MOTS-C	1.2/2mg	SEMI-SYN	Injection site reaction, mild nausea (rare)	Fat oxidation + mitochondrial metabolism
AOD-9604 + MOTS-C + Tesamorelin	1.2/2/3mg	SEMI-SYN	Fluid retention, joint pain, injection site effects	Full body recomposition stack
BPC-157 + TB-500	3/3mg	SEMI-SYN	Mild fatigue, dizziness, transient headache	Accelerated tissue & injury repair
BPC-157 + KPV + TB-500	3/3/3mg	SEMI-SYN	Nausea, mild lethargy, injection site reactions	Repair + gut inflammation + healing
BPC-157 + GHK-Cu + KPV + TB-500	3/10/3/3mg	SEMI-SYN	Flushing, injection site irritation, mild fatigue	Full-spectrum regenerative protocol
CJC-1295 + Ipamorelin	1.2/2mg	SYNTHETIC	Water retention, tingling, mild hunger, headache	GH pulse optimization; body composition
Tesamorelin + Ipamorelin	3/2mg	SEMI-SYN	Fluid retention, joint pain, injection site pain	Visceral fat + GH secretagogue synergy
DSIP + BPC-157 + CJC-1295	1/2/2mg	SEMI-SYN	Drowsiness, headache, mild fluid retention	Sleep, repair & GH — recovery stack
GHK-Cu + Epithalon	10/2mg	SEMI-SYN	Injection site reactions, transient fatigue	Skin rejuvenation + telomere anti-aging
SEMAX + SELANK	1/1mg	SEMI-SYN	Mild headache, nasal irritation (intranasal), fatigue	Cognitive clarity + anxiolytic balance
Pinealon + PE22-28 + SELANK	2/2/2mg	SYNTHETIC	Transient headache, dizziness, drowsiness	Neuroprotection + memory + calm focus

Oral Capsule Formulations

Product	Dose	Nature	Common Side Effects	Indication / Use
BPC-157	500mcg capsule	SEMI-SYN	Mild nausea, GI discomfort (rare at oral doses)	Gut healing & systemic anti-inflammatory
Ondansetron	4mg tablet	SYNTHETIC	Headache, constipation, QT prolongation (rare)	Anti-nausea; GLP-1 side-effect management
SLU-PP 332	250mcg capsule	SYNTHETIC	Limited human data; potential GI upset, insomnia	Exercise mimetic; mitochondrial biogenesis
5-Amino (5-AMP)	50mg capsule	NATURAL	Mild GI upset, fatigue at higher doses	AMPK activator; metabolic & cellular energy
Ibutamoren (MK-677)	25mg capsule	SYNTHETIC	Increased appetite, water retention, fatigue, numbness	Oral GH secretagogue; deep sleep; recovery
Dihexa	5mg capsule	SYNTHETIC	Headache, irritability, vivid dreams (limited human data)	Nootropic; synaptic formation; memory & learning
SLU-PP 332 + BAM	100mcg/15mg	SYNTHETIC	GI discomfort, potential CNS stimulation	Dual metabolic activator combo

Product	Dose	Nature	Common Side Effects	Indication / Use
Dihexa + Tesofensine	5mg/500mcg	SYNTHETIC	Elevated HR/BP, dry mouth, insomnia, nausea	Cognitive + appetite control synergy

Ancillary & Support Agents

Agent	Concentration	Nature	Common Side Effects	Clinical Application
NAD+ (1200mg)	200mg/mL • 6mL	NATURAL	Flushing, nausea, chest tightness, headache (rate-dependent)	IV/IM; bioidentical coenzyme; energy, detox, longevity
Glutathione (2000mg)	200mg/mL • 10mL	NATURAL	Skin lightening with frequent IV use, abdominal cramping	IV push antioxidant; liver, immune, skin brightening
Methylcobalamin B12 (10mg)	1mg/mL • 10mL	NATURAL	Generally very well tolerated; rare acne, mild diarrhea	Bioidentical active B12; neuro support; methylation
Sermorelin (15mg)	2.5mg/mL • 6mL	SEMI-SYN	Injection site pain, flushing, headache, fluid retention	Synthetic GHRH analog; anti-aging; GH stimulation

■ **IMPORTANT NOTICE:** This document is intended for licensed healthcare practitioners only. All compounded medications require a valid patient-practitioner relationship and appropriate clinical evaluation. Side effects listed are the most commonly reported; this is not an exhaustive safety profile. These products have not been evaluated or approved by the FDA for the specific indications listed. 503A products require individual patient prescriptions. 503B products may be distributed to registered healthcare facilities per federal guidelines. 'Natural' classifications refer to structural origin only and do not imply greater safety or efficacy. For professional use only — not for retail distribution.